FinForge AI

Autonomous Financial Filing Auditor & Certified Verification Engine

EXECUTIVE BRIEF

PRODUCT	CORE ARCHITECTURE	GROUND TRUTH	LIVE DEPLOYMENT
FinForge AI v2.4	LangGraph HITL + FastAPI	SEC EDGAR XBRL Data	finforge-ten.vercel.app

1. Executive Summary & Mission Statement

FinForge AI is an enterprise financial auditing system built to address the critical risks of autonomous AI hallucination in corporate financial filings. When processing massive SEC Form 10-K disclosures, generic Large Language Models frequently misinterpret accounting line items, confuse GAAP with non-GAAP definitions, or hallucinate metrics without verifiable source groundings.

FinForge enforces mathematical double-entry accounting invariants, cross-checks extractions against authoritative SEC EDGAR XBRL taxonomies, and features a stateful **Human-in-the-Loop (HITL)** escalation gateway. When discrepancies or low-confidence metrics occur, autonomous execution pauses deterministically, allowing certified human auditors to review exact 10-K evidence excerpts and **Approve**, **Correct**, or **Reject** values before certified audit certificates are issued.

> **Key Value Metric**
>
> FinForge maintains strict append-only non-destructive versioning: raw autonomous extractions are **never overwritten**. Every auditor modification is immutably recorded in a SHA-256 tamper-evident hash ledger.

2. System Architecture & Technical Stack

The system follows a decoupled, resilient architecture designed for both high-throughput serverless cloud execution and secure on-premise local deployments:

- **Frontend Application**: Modern Single-Page Application (SPA) built with React 18, Tailwind CSS, JetBrains Mono typography, inline PDF certificate renderer (`html2pdf.js`), and responsive financial KPI monitors.
- **Backend Application Layer**: Python FastAPI asynchronous API service supporting all 7 Hackathon Core REST specifications alongside 12 advanced financial analytics endpoints.
- **LangGraph Workflow Engine**: 9-stage deterministic Directed Acyclic Graph (DAG) with state checkpointing, thread pausing, and reactive resumption.
- **Data Persistence**: SQLite append-only datastore tracking `audit_runs`, `extracted_metrics`, `review_items`, `audit_history`, and SHA-256 `audit_trail_blocks`.

3. The 9-Stage Autonomous Audit Workflow

1 **SEC Ingestion**

Automated retrieval of Form 10-K annual reports or custom filing uploads.

2 **Context Chunking**

Segmenting Item 8 Consolidated Balance Sheets and Footnotes.

3 **Fact Extraction**

Targeted extraction of Assets, Liabilities, Equity, Revenue, and Income.

4 **Invariant Verification**

Mathematical check: Total Assets = Total Liabilities + Stockholders' Equity.

5 **Temporal Analytics**

YoY variance checks against multi-year historical company baselines.

6 **Adaptive Scoring**

Dynamic confidence scoring (0.00 to 1.00) based on OCR & math clarity.

7 **Escalation Gate**

Halts thread if confidence < 85% or accounting invariants fail.

8 **Human Review**

Auditor reviews evidence and issues Approve / Correct / Reject decision.

9 **Reconciliation & Seal**

Workflow resumes, validates against XBRL, and issues certified PDF report.

4. Realistic Anomaly Demonstration: NVIDIA (NVDA FY2023)

To validate FinForge under real-world filing conditions, an intentional accounting anomaly was injected into the NVIDIA Corp FY2023 Form 10-K balance sheet data:

Financial Metric	Extracted Value	SEC EDGAR Ground Truth	Variance	Confidence	System Status
Total Revenue	$26,974.0M	$26,974.0M	$0.0M	98%	Passed
Net Income	$4,368.0M	$4,368.0M	$0.0M	96%	Passed
Total Liabilities	$27,241.0M	$27,241.0M	$0.0M	95%	Passed
Stockholders' Equity	$13,941.0M	$13,941.0M	$0.0M	95%	Passed
Total Assets (Anomalous)	**$45,182.0M**	**$41,182.0M**	**+$4,000.0M**	68%	⚠️ Flagged for Review

> **Autonomous Error Detection Logic**
>
> FinForge detected that $ ext{Total Liabilities (\$27,241M)} + ext{Equity (\$13,941M)} = \mathbf{\$41,182M}$, leaving a **$4,000.0M discrepancy** against the extracted Total Assets ($45,182M). The execution thread was automatically paused at the `human_review` stage.

Auditor Decision & Certification Resolution:

1. The auditor opens the **Human Review Center**, which highlights the $4,000.0M variance alongside the original SEC Form 10-K evidence text.
2. The auditor clicks **Correct to $41,182.0M** (or selects Custom Justification).
3. FinForge writes the correction to `audit_history`, marks the metric as passed, and resumes the LangGraph thread to `completed`.
4. The final reconciliation matrix updates to **100.0% extraction accuracy** with zero outstanding discrepancies.

5. REST API Contract & Core Endpoints

Method	Endpoint	Description
`GET`	`/api/companies`	Returns supported companies and current filing audit readiness.
`POST`	`/api/audit/run`	Initiates 9-stage audit for a given ticker and reporting period.
`POST`	`/api/audit/upload`	Uploads and ingests custom Form 10-K filing extractions.
`GET`	`/api/audit/{id}`	Returns detailed audit execution state, stage, and metric results.
`GET`	`/api/reviews`	Lists all items currently paused and awaiting human review.

Method	Endpoint	Description
POST	`/api/reviews/{id}/decision`	Submits auditor decision (Approve, Correct, Reject) and resumes LangGraph.
GET	`/api/audit/{id}/report`	Retrieves official SEC XBRL ground truth reconciliation report.
GET	`/api/audit/{id}/history`	Retrieves immutable append-only decision audit trail.
POST	`/api/audit/reset`	Clears all previous runs and reviews for a clean demonstration run.

6. Conclusion & Verification Summary

FinForge AI demonstrates how autonomous AI pipelines can be safely deployed in high-stakes financial domains. By combining automated extraction speed with strict accounting invariant enforcement, cryptographic auditing, and Human-in-the-Loop decision controls, FinForge provides a trustworthy, audit-ready standard for enterprise financial analysis.